

SRIRATHI MUTHURAMAN

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I am a PhD student in neuroscience investigating the neural mechanisms underlying motor and cognitive dysfunctions in neuropsychiatric disorders. Experienced in conducting independent research, troubleshooting experiments, and interpreting complex datasets. Interested in pursuing an industrial R&D role where I can apply my scientific and analytical skills to biomedical and therapeutic research.

EDUCATION

2023 – ONGOING

DOCTORAL STUDENT, NEUROSCIENCE

DEPARTMENT OF PHYSIOLOGY, UNIVERSITY OF TARTU, ESTONIA

2021 – 2023

MASTER OF TECHNOLOGY, BIOTECHNOLOGY

B. S. ABDUR RAHMAN INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI, INDIA

CGPA – 9.43

2017 – 2021

BACHELOR OF TECHNOLOGY, BIOTECHNOLOGY

DR. MGR EDUCATIONAL AND RESEARCH INSTITUTE, CHENNAI, INDIA

CGPA – 9.51

DOCTORAL RESEARCH PROJECT

Proteomic Brain Endophenotypes and Molecular Correlates of Motor Phenotype Alterations in *Negr1* Deficient Mice (2023 – Ongoing)

- Investigate molecular mechanisms underlying motor and cognitive dysfunction using genetically modified mouse model.
- Perform rodent behavioural experiments to characterise motor and functional phenotypes.
- Perform mouse dissection and transcardial perfusion for downstream molecular and histological analyses.
- Conduct qPCR and gene expression studies and perform protein-based experiments including protein extraction, Western blotting, kinase activity assays, and immunostaining.
- Analyse mass spectrometry/proteomics datasets and perform gene ontology analysis to identify biological pathways associated with experimental phenotypes.
- Use GraphPad Prism and R for statistical analysis and interpretation of experimental data.

TECHNICAL EXPERTISE

Molecular Biology

Isolation of DNA, RNA and protein, qPCR, Molecular cloning, Plasmid construction

Biochemistry

Protein isolation, SDS PAGE, Western Blot, Kinase activity assay, Immunostaining

Animal Studies

Zebrafish - Maintenance, Breeding, Dechoriation, Toxicology studies, Microinjection

Mouse – Dissection, Behavioural experiments, Transcardial perfusion

Statistical Analysis

GraphPad Prism, R programming

CONFERENCES AND WORKSHOPS

- UK Theoretical Neuroscience Workshop, City St George's, University of London, UK (16.12.2025) **POSTER PRESENTATION**
- Oxford Autumn School in Neuroscience 2025, University of Oxford, England, UK (9-10th October 2025)
- TEATIME Final Conference – “The TEATIME Journey: Shaping the Future of Lab Animal Monitoring” in Helsinki, Finland (2-3rd September 2025) **POSTER PRESENTATION**
- Summer School in Brain Disorders Research at Cardiff University's Centre for Neuropsychiatric Genetics & Genomics, Wales, UK (14- 17th July 2025) **POSTER PRESENTATION**
- Second TEATIME Beginners Training School in Bratislava, Slovakia (7-20th June 2025)
- TEATIME Experimental Design course (April 2025)
- Multiomics Approaches in Reproductive Medicine – from the Wet Lab to the Data Analysis course at CELVIA Academy in Tartu, Estonia (20-21st March 2025)
- Training camp at Voore, The Art of Giving a Popular Science Talk, and participated in the oral presentation (10-11th May 2024) **ORAL PRESENTATION**
- Competence Course in Laboratory Animal Science at the Estonian University of Life Sciences (25.03.2024 - 10.05.2024)
- DIGS-BB Summer school – Biological making of materials, Technische Universität Dresden, Germany (11-15th September 2023)

PUBLICATION

- **Psychomotor disturbances in Neuronal growth regulator-1 deficient (*Negr1*^{-/-}) mice are associated with sex-specific alterations in hippocampal and skeletal muscle signaling (2026 – Under review)**
Srirathi Muthuraman, Mohan Jayaram, Darja Lavogina, Birgit Pilve, Arpana Reinsberg, Kaie Mikheim, Eero Vasar, Mari-Anne Philips, Katayani Singh
- **Hippocampus-focused glutathione-dependent redox alterations in male *Negr1*^{-/-} mice (2026 – Under review)**
Narges Mohammadrahimi, Este Leidmaa, Darja Lavõgina, Maria Kaare, **Srirathi Muthuraman**, Rando Porosk, Olga Bulenko, Marten Kase, Johanna Karp, Denisa Nicoleta Vicol, Indrek Heinla, Katayani Singh, Stefania Schiavone, Eero Vasar, Mari-Anne Philips

- **Sex-specific changes in the hippocampal proteome of *Negr1*^{-/-} mice: insight into the mechanisms of neuropsychiatric disorders (2026)**
Srirathi Muthuraman, Mohan Jayaram, Liisi Promet, Toomas Jagomäe, Arun Kumar Devarajan, Andreas-Christian Hade, Mari-Anne Philips, Katyayani Singh & Eero Vasar
- **The IgLON family of cell adhesion molecules expressed in developing neural circuits ensure the proper functioning of the sensory system in mice (2024)**
Katyayani Singh, Mohan Jayaram, Arpana Hanumantharaju, Tambet Tõnissoo, Toomas Jagomäe, Kaie Mikheim, **Srirathi Muthuraman**, Scott F. Gilbert, Mario Plaas, Michael K. E. Schäfer, Jürgen Innos, Kersti Lilleväli, Mari-Anne Philips & Eero Vasar

RESEARCH SUPERVISION & MENTORING

Supervised a bachelor's research project focused on '**Molecular Analysis of Skeletal Muscle in *Negr1* Deficient Mice**' (September 2025 - June 2026)

- Guided experimental planning, execution, data analysis, and interpretation.
- Trained and mentored the student in RNA extraction, qPCR and statistical analysis.
- Provided regular feedback on experimental progress and scientific presentation.

ACHIEVEMENTS

Mobility Grant for Doctoral Students, Faculty of Medicine (2026)

Awarded a mobility grant to support a research internship at the Myofin Research Lab, Folkhälsan Research Center, Helsinki, Finland.

Mobility Grant for Doctoral Students, Faculty of Medicine (2025)

Awarded a mobility grant to support participation in the UK Theoretical Neuroscience Workshop at City St George's, University of London, UK.

NewGen IEDC Seed Funding Award (2020)

Selected under the NewGen Innovation and Entrepreneurship Development Centre (NewGen IEDC) scheme for the bachelor's dissertation project "Probiotic Instant Millet Drink – An Advancement in Millet Food Preservation" and awarded INR 2,50,000 in seed funding by the Department of Science and Technology (DST), Government of India.

LANGUAGE SKILLS

Tamil — Native language

English — Fluent (C1 Advanced)

Estonian — A1.2 (Basic proficiency)

DECLARATION

I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief.

Place: Tartu, Estonia

Date: 04.08.2026



Signature

(SRIRATHI MUTHURAMAN)